



Assignment 1 – Fall 2025

Course Title:	Internet of Things	Course Code:	CPE 445	Credit Hours:	3(3,0)
Course Instructor:	Dr. Abbas Javed	Programme Name:			
Semester:		Batch:		Section:	FA22-BCE-A & B
Submission Date:	19/10/2025	Maximum Marks:			10
Name:		Registration Number:			
Important Instructions / Guidelines:					
In case of plagiarism, you will be marked ZERO.					

Question 1: (CLO1- C2)

[10]

An IoT based company is planning to install smart streetlights in Lahore, Pakistan. Smart streetlight has to exhibit following features:

1. Monitor the daylight.
 2. Based on daylight switch ON/OFF the LED light.
 3. If there is no motion detection turn OFF the light after 1 minute.
 4. In case of accident, the smart streetlight should navigate the emergency responders to the location of accident by blinking streetlights
 5. Traffic management through real-time data feeds that track congestion and speed
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- a) As a design engineer, you need to explain the IOTWF architecture for smart streetlights. (3)
 - b) Explain the design of smart streetlight that how you will able to implement features (1-4) (3)
 - c) Summarize your choice for sensors, actuators and access technology by considering limitations of constrained nodes and constrained networks. (4)